UC San Diego

Deep Learning – Experiment Tracking

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MLOps

- Machine Learning Operations - MLOps
  - ML + DevOps
  - Practices & tools to streamline and automate deployment, monitoring, and management of ML models
- Key components
  - Data management - data versioning, data validation, tracking lineage
  - Model training - experiment tracking, automate pipelines
  - Deployment - model registry, model deployment and serving
  - Monitoring - monitoring model performance, accuracy drift, alerts

Experiment Tracking

- Logging metadata about experiments
  - hyperparameters
  - data version
  - model/code version
- Training metrics
  - loss, accuracy, …
  - artifacts: model weights, training logs, …
- System
  - memory usage
  - compute utilization

MLflow Vs. Weights & Biases

MLflow

- Self-host or managed
- Open-source
- Free; pay for infrastructure/managed
- Functional, simple UI
- Full ML lifecycle
- Good for teams prioritizing self-hosted, end-to-end control

Weights & Biases

- Cloud or privately hosted
- Open-source Client/SDK with SaaS platform
- Free tier; paid for teams
- Polished, interactive UI
- Experiment tracking & collaboration
- Good for teams prioritizing visualization & collaboration

Weights & Biases

- A comprehensive MLOps platform for tracking, visualizing, and managing machine learning experiments
- Key Features
  - Experiment Tracking
    - Log metrics, hyperparameters, system stats (GPU/CPU), and plots in real-time
  - Model & Data Versioning
    - Track model checkpoints and datasets using Artifacts for full lineage tracking
  - Hyperparameter Sweeps
    - Automate complex optimization tasks to find the best model architecture quickly.

MLFlow

- MLFlow lets you manage the end-to-end ML Lifecycle
- Essential features:
  - MLflow Tracking: Log/compare experiments, parameters, and results
  - MLflow Projects: Package code for reproducible runs
  - MLflow Models: Deploy models to diverse platforms
- Open source, library-agnostic, and collaborative
- Can be self-hosted or hosted on Databricks' servers

Demos

Wandb & MLflow - Comparing ML Models

- Code
  - mlops/mlflow_demo.ipynb
  - mlops/wandb_demo.ipynb
- Description
  - Airline delay classification: Predict whether a flight will be delayed
  - Compare models:
    - Logistic Regression, Decision Tree, Random Forest, XGBoost
  - Using MLflow or W&B to track and compare runs

Weights & Biases Setup

- Set up an account with Weights & Biases - https://wandb.ai/site/
- Install Weights & Biases:
  - pip install wandb (No need to do this for the demo as the Singularity container already has W&B)
- Run this command in the terminal:
  - wandb login
  - This will prompt you to paste an API key. You can get it from:
    - https://wandb.ai/authorize
- Once logged in, your credentials will be saved in your system. Future runs will be auto-authenticated without re-entering the key.

WandB Setup

- Set up an account with Weights & Biases
  - https://wandb.ai/site/
- Run command in terminal on compute server:
  - `wandb login`
- Paste your API key
  - Login command will prompt yout to paste your WandB API key, which can be retrieved from https://wandb.ai/authorize
  - After first login, your credentials will be saved to your system (~/.netrc). Future runs will be auto-authenticated without having to re-enter your API key.
  - Default port is 5000. Can change to different port number if needed.

MLFlow Setup

- In the terminal, make sure you're in the same directory as the launch_mlflow.sh file, then run the following
  - bash launch_mlflow.sh
- The launch_mlflow.sh script starts an MLflow server on a random available port and connects it to Expanse's reverse proxy service, giving you a unique public URL to access the MLflow UI in your browser every time.
  - It should look like this
    - https://<to-be-replaced>.expanse-user-content.sdsc.edu
- The server will keep running until your job ends, at which point your run data is saved automatically.

MLFlow Setup

- In the notebook, set the tracking URI to be the generated link
  - mlflow.set_tracking_uri('https://<to-be-replaced>.expanse-user-content.sdsc.edu')
- This allows MLflow to log runs to the generated link.
- In the UI
  - Navigate to the experiment.
  - Make sure you're on Model Training and not GenAI in the sidebar.
  - Then click on any run.
  - You can see graphs with your experiment metrics under the `Model Metrics` tab

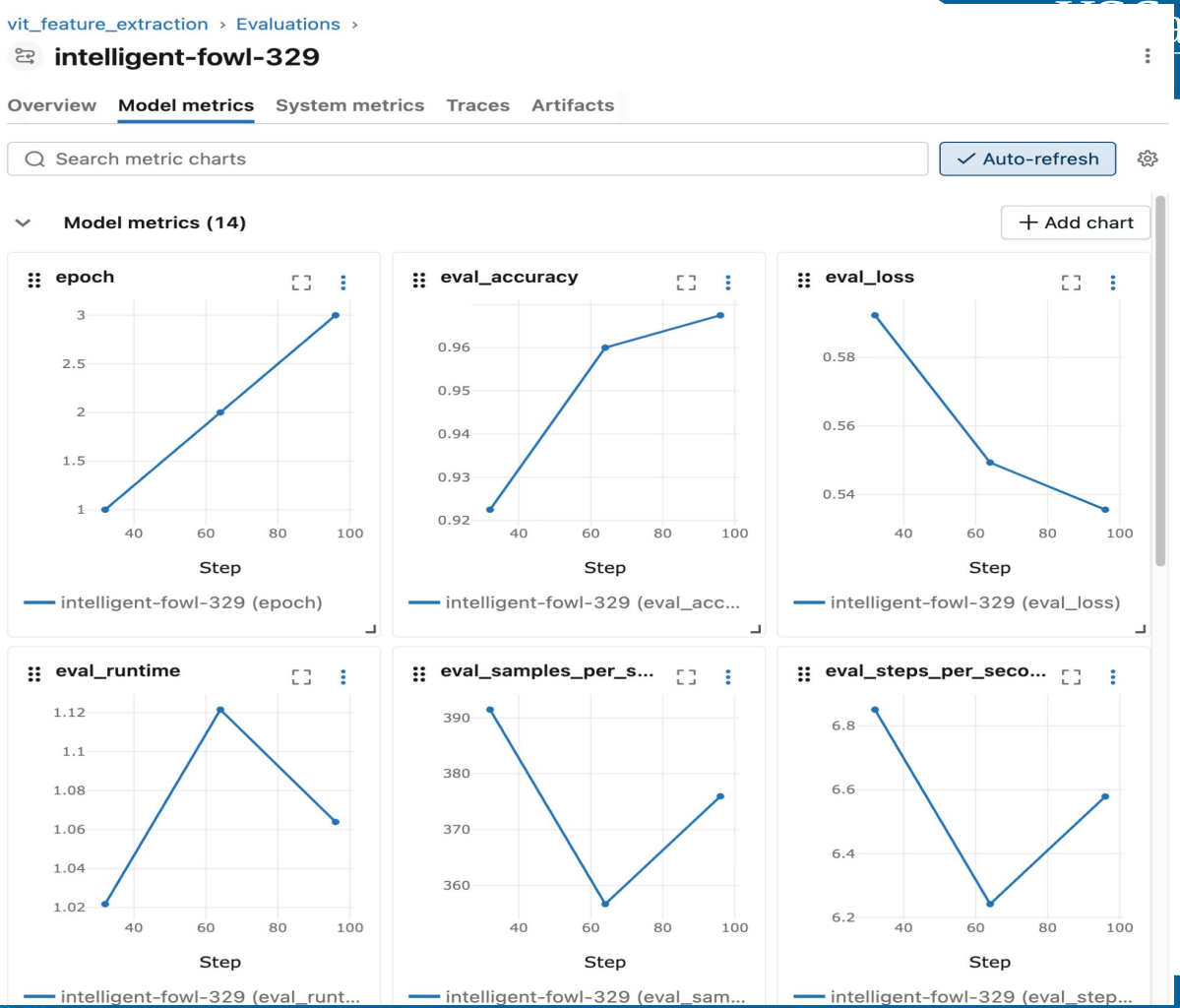
Finetuning with Logging

- Code
  - Huggingface/logger_finetuning_hf.ipynb
- Description
  - Task : Classify cats vs dogs
  - Approach: Transfer learning using finetuning
  - Model: HuggingFace Vision Transformer (ViT)
  - Logging:
    - Training metrics (e.g., loss, accuracy) logged to both W&B and MLFlow
    - System metrics (e.g., GPU utilization) can be cleanly seen in W&B

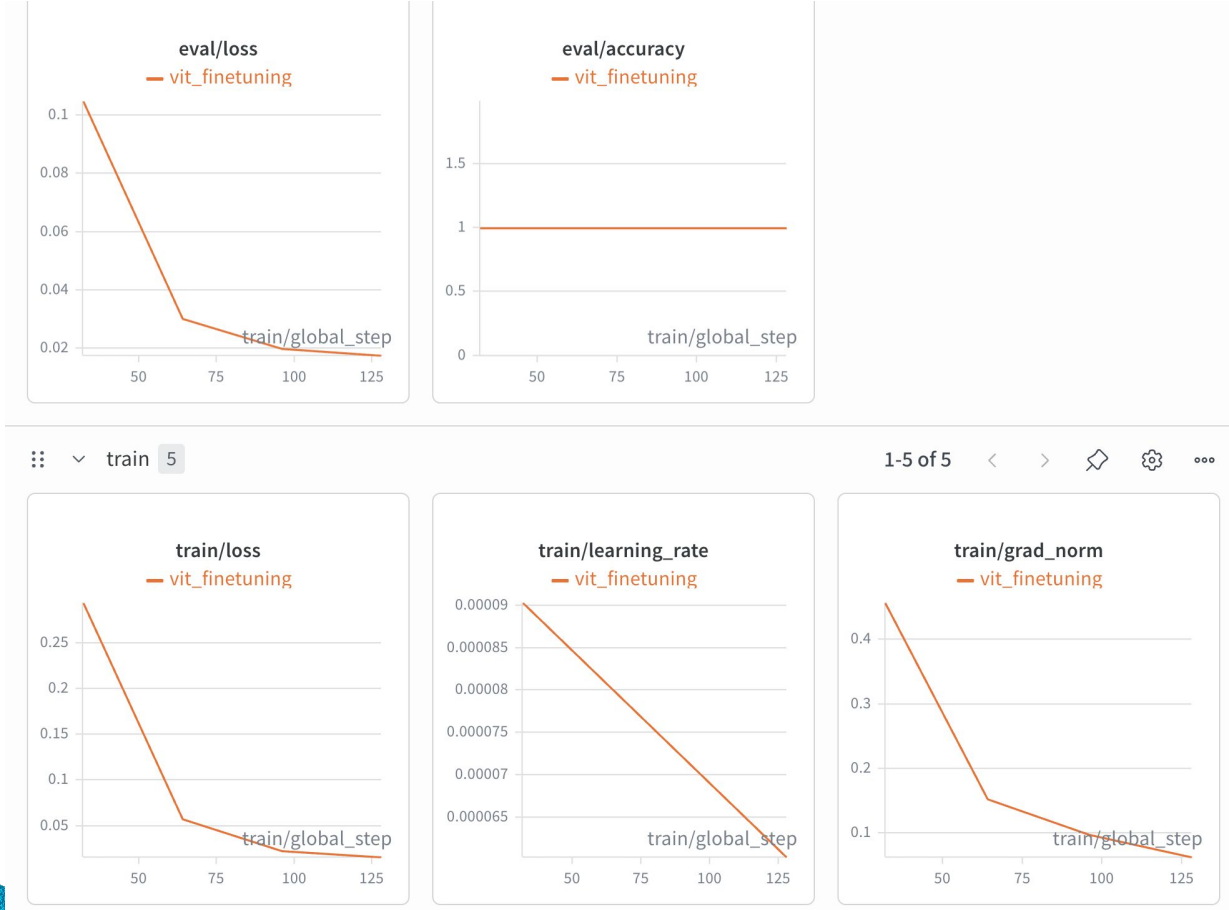
W&B Hyperparameter Sweep

- Code
  - Huggingface/finetuning_hf_sweep_wandb.ipynb
- Description
  - Task : Classify cats vs dogs
  - Approach: Transfer learning using finetuning
  - Model: HuggingFace Vision Transformer (ViT)
  - Logging: W&B sweep agent conducts a hyperparameter sweep over a defined sweep configuration
    - Using Bayesian hyperparameter optimization

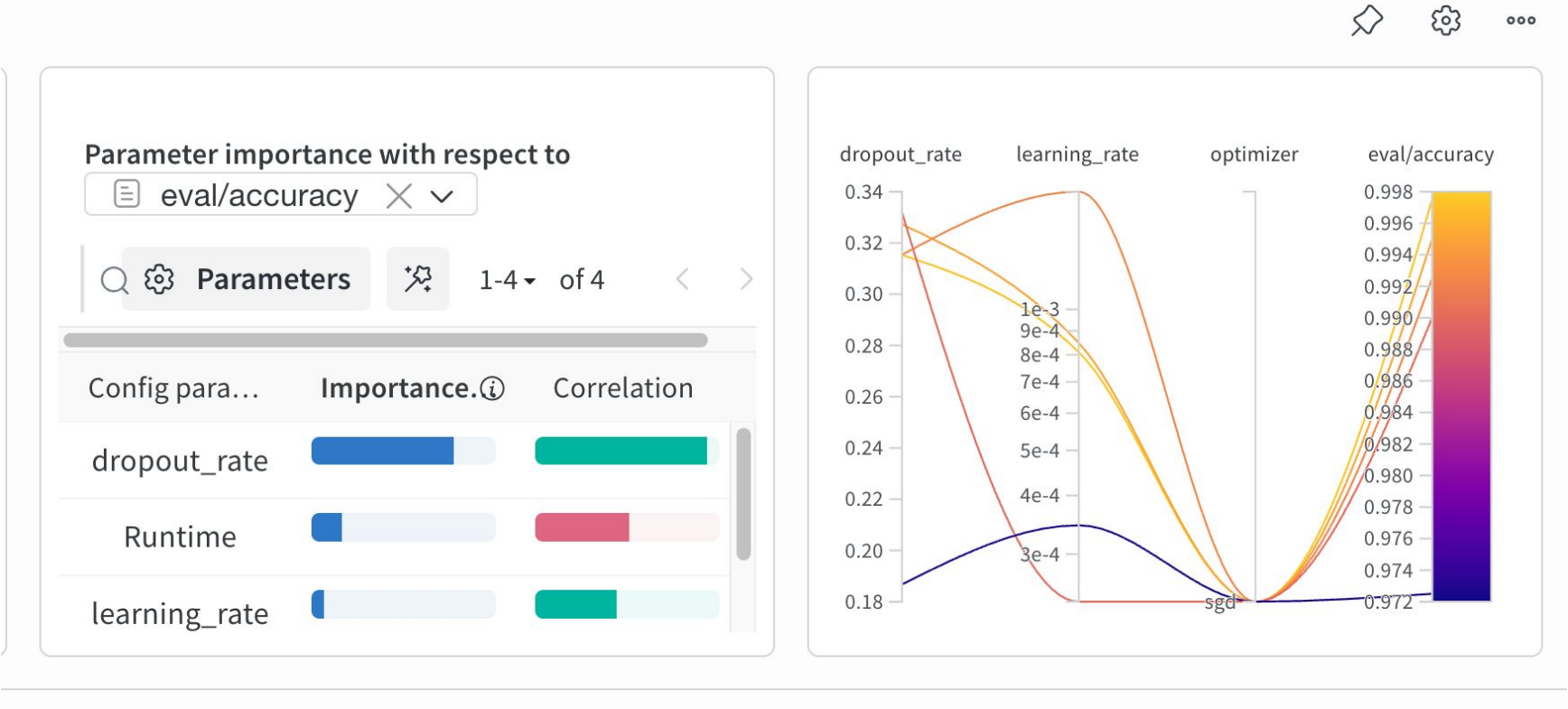
MLflow Chart



W&B Charts


eval/loss
vit_finetuning
0.1
0.08
0.06
0.04
0.02
train/global_step
50
75
100
125
eval/accuracy
vit_finetuning
1.5
1
0.5
0
train/global_step
50
75
100
125
train 5
1-5 of 5
train/loss
vit_finetuning
0.25
0.2
0.15
0.1
0.05
train/global_step
50
75
100
125
train/learning_rate
vit_finetuning
0.00009
0.000085
0.00008
0.000075
0.00007
0.000065
train/global_step
50
75
100
125
train/grad_norm
vit_finetuning
0.4
0.3
0.2
0.1
train/global_step
50
75
100
125

W&B Sweep


Parameter importance with respect to
eval/accuracy
Parameters
1-4
of 4
Config para…
Importance.
Correlation
dropout_rate
Runtime
learning_rate
dropout_rate
learning_rate
optimizer
eval/accuracy
0.34
0.32
0.30
0.28
0.26
0.24
0.22
0.20
0.18
1e-3
9e-4
8e-4
7e-4
6e-4
5e-4
4e-4
3e-4
sgd
0.998
0.996
0.994
0.992
0.990
0.988
0.986
0.984
0.982
0.980
0.978
0.976
0.974
0.972